

Supporting Teens' Intentional Social Media Use Through Interaction Design

An exploratory proof-of-concept study

Katie Davis
University of Washington, USA
kdavis78@uw.edu

Petr Slovak
Department of Informatics, King's College London, UK
petr.slovak@kcl.ac.uk

Rotem Landesman
Information School, University of Washington, USA
roteml@uw.edu

Caroline Pitt
University of Washington, USA
pittc@uw.edu

Abdullatif Ghajar
Hasso Plattner Institute, University of Potsdam, Potsdam, Germany
Abdullatif.ghajar@student.hpi.uni-potsdam.de

Jessica Lee Schleider
Department of Psychology, Stony Brook University, USA
jessica.schleider@gmail.com

Saba Kawas
Computer Science and Engineering, University of Minnesota, USA
skawas@umn.edu

Andrea Guadalupe Perez Portillo
School of Social Work, University of Washington, USA
andreagp@uw.edu

Nicole S. Kuhn
University of Washington, USA
nskuhn06@uw.edu

ABSTRACT

Prior research indicates that many adolescents struggle to use social media intentionally, with negative impacts on well-being. This presents an important but under-researched opportunity to improve youth well-being through interaction design. We developed Locus, a mobile application that shapes adolescents' entry experiences into the social media apps on their phones by prompting them to reflect on their social media intentions before they open an app and at the end of the day. In designing Locus, we combined psychology theories—to identify design targets—with co-design approaches—to design around these targets for youth's lived experiences. Following co-design work with nine teens (M_{age}=15.7 years), we conducted an open trial field deployment (N = 54; M_{age}=16.2 years) to explore how Locus was used and experienced by adolescents during their daily lives. Results provide preliminary evidence that in-the-moment interventions for self-regulation skills can support adolescents' intentional social media use.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

KEYWORDS

digital well-being, adolescents, social media, intervention design



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1 INTRODUCTION

As many as 25% of adolescents engage in problematic social media use (PSMU) [49, 72], characterized by difficulty controlling social media behavior, with negative impacts on sleep, academic performance, relationships, and well-being [3]. Even among adolescents whose behavior does not meet PSMU criteria, prior work indicates that it is common for youth to approach their social media use in an unintentional way, mindlessly scrolling through their feeds, allowing auto-play to show them video after video, and reacting to rather than interacting with the people and content they encounter online [20, 81]. Youth report experiencing these practices as unsatisfying but say they struggle to break their habits [20, 81]. Unintentional social media use is associated with lowered levels of subjective well-being among youth [17, 60].

We developed *Locus*,[1] a mobile application that shapes adolescents' entry experiences into the social media apps on their phones by prompting them to reflect on their social media intentions before they start a session. Locus also delivers end-of-day reflection prompts, sent as push notifications to a user's home screen, that encourage adolescents to reflect on their social media use that day

[1]The name Locus comes from the psychological construct 'locus of control' [69]. People with a strong *internal* locus of control have a greater sense of personal agency; people who tend toward an *external* locus of control are more likely to feel that external factors such as chance or luck control their experiences. With Locus, we aim to support adolescents in shifting from an external to an internal locus of control in their social media use.

and set goals for the next day. To design Locus, we used psychology theories to identify design targets and combined them with co-design approaches to design around these targets for youth's lived experiences. We then conducted an open trial field deployment ($N = 54$; M_{age}=16.2 years) to explore the Locus intervention's acceptability and uptake, and to understand how it was experienced by adolescents during the course of their daily lives.

The results of our quantitative analysis showed that, on average, adolescents' self-control and autonomy in relation to their social media use increased over the two week-period of using Locus, while their levels of absentminded social media use decreased. In follow-up interviews, participants said that Locus helped them to engage in self-regulation behaviors around their social media use, which they said led to a greater sense of meaning and agency in their social media interactions. The Locus design and field deployment results contribute new insight into the acceptability and potential effectiveness of design mechanisms that support adolescents' intentional social media use.

2 RELATED WORK

2.1 Well-Being, Agency, and Self-Regulation in Adolescents' Social Media Use

Although many adolescents derive benefits from their social media engagement (e.g., [17, 19, 20, 66, 81]), as many as 25% experience problematic social media use (PSMU) [3, 5, 17, 37, 49, 72]. PSMU is sometimes encompassed within the more general construct of problematic internet use (PIU) and is related to problematic smartphone use (PSU) [3]. Core to definitions of PSMU, PIU, and PSU are difficulties controlling one's behavior in relation to technology use, with negative impacts on sleep quality, relationships, academic performance, and personal well-being [3, 24, 37, 49].

Even among those adolescents whose social media use does not meet PSMU criteria, many youth nevertheless struggle to maintain a sense of purpose and intention when they embark on a social media session. Weinstein and James [81] documented the many pulls on adolescents' attention, including alluring personalized content, infinite scroll, the unpredictable rewards and social validation associated with features such as the like button and comments, and the "quicksand" of social media-based social comparisons (see also [54–56]). Adolescents who fall down these social media "rabbit holes" are more likely to report lowered subjective well-being [56, 60].

Although people of all ages experience agency-undermining experiences with social media (e.g., [42, 78]), adolescents may find it particularly difficult to achieve a sense of personal agency on TikTok, Snapchat, and other social media platforms. This is due to the gap between adolescents' heightened emotional responses and their lagging self-regulation skills. Compared to children and adults, adolescents display an increased sensitivity to social situations and a greater tendency toward emotional reactivity when confronted with such situations [74, 75]. At the same time, self-regulation skills such as reflection, planning, goal-setting, and self-monitoring [4, 7, 51] are relatively under-developed during this developmental period [9, 10]. Indeed, Van Deursen et al. [22] found that adolescents were more likely than adults to have trouble engaging in self-regulation behaviors when using their smartphones.

To summarize, the fundamentally social nature of social media, combined with persuasive designs such as infinite scroll and notifications, regularly place adolescents in situations where their tendency toward emotional reactivity overwhelms their still developing self-regulation abilities [20]. This body of work points to the value of intervening in adolescents' social media use at the moment when self-regulation skills are most needed and least likely to be used.

2.2 Interventions to Support Digital Agency and Well-Being

Smartphone overuse is a prevalent target of HCI interventions that focus on designing for personal agency and well-being. *MyTime* [29] supports people in setting and achieving goals related to the time they spend on their smartphones. *TypeOut* [84] combines a typing-based unlock process with self-affirmation content to support users in reducing smartphone overuse. Both interventions target smartphone overuse by supporting people's sense of personal agency. Other interventions targeting smartphone reduction similarly focus on promoting personal agency and well-being (e.g., [33, 38, 40, 57]).

Increasingly, interventions are targeting the quality of phone use in addition to or instead of quantity of use as a core outcome (recognizing that these two dimensions are often entwined). For instance, *Chirp* [86] is a mobile Twitter client that researchers developed to increase users' sense of agency during their Twitter use. *MindPhone* [77] draws on mindfulness techniques to target absentminded smartphone. The intervention's focus on goal-setting and reflection align it with other interventions that target self-regulation skills as a mechanism for supporting personal agency in phone use (e.g., [34, 39, 45, 48]). In the current study, we build on this work by focusing on self-regulation skills as the primary mechanism for promoting adolescents' sense of agency in their phone-based social media use.

When considered alongside theoretical work (e.g., [43, 45]) and empirical investigations of commercial technologies (e.g., [16, 46, 47]), existing interventions point to some overarching takeaways for future research focused on designing for personal agency and well-being. Overly restrictive techniques such as lockout mechanisms, while sometimes effective, can lead to user frustration and even an increase in phone usage [35, 46, 47]. Notification messages and goal-setting features are less restrictive, but some researchers caution they can be easily dismissed [41, 84]. Lyngs et al. [46] propose a "Goldilocks" level of support for digital self-control interventions that target self-regulation. Such design mechanisms should adapt to users' personal goals and usage patterns and provide enough support to encourage behavior change without being so restrictive that users abandon the intervention altogether.

A notable gap in this research base is the scarcity of interventions that are designed from the beginning with adolescent users in mind [28, 68]. In their recent meta-analysis of digital self-control tools, Roffarello and De Russis found that only two of the 64 studies they reviewed included youth of any age; the vast majority of studies involve university students and office workers. Outside of HCI, teen-focused interventions have been developed that target self-regulation in the context of problematic technology use (e.g.,

[32, 36, 72, 79, 80]). However, as school-based interventions [32] or clinician-directed exercises [80], these interventions tend to be temporally removed from the situations for which the target skills are needed, posing a known challenge in skill transfer from intervention to everyday use [73].

Within HCI, research is starting to explore interaction designs that support adolescent well-being and address the challenge of skill transfer [18]. Some of this work addresses specific areas, such as cyberbullying (e.g., [23, 26]), online safety (e.g., [2, 59, 83]), sexual health [63], and physical health [82], whereas other interventions target adolescent mental health and well-being more generally (e.g., [8, 61, 62, 64, 67]). Chen et al. [15] designed a technology probe for adolescents undergoing psychiatric treatment for smartphone addiction that includes self-monitoring and parental feedback mechanisms. Results from tracking seven parent-teen dyads over two weeks point to the promise of using interaction designs to support adolescent self-regulation in the context of their technology use.

To the best of our knowledge, there is no published work examining interventions that provide in-the-moment support for PSMU in adolescents. The closest work is Hou et al. [30], who sought to support adolescents' self-regulation behaviors around their social media use by asking study participants to record and send researchers daily reflections about their social media use. However, because this intervention centered daily communication with a researcher, it did not function as a standalone intervention that is embedded directly into adolescents' daily social media use. There is a need to develop and test intervention mechanisms that make use of mobile technologies to reach adolescents where they are and when they are most in need of intervention [1, 28, 52, 71].

3 LOCUS APP DESIGN PROCESS

3.1 Identifying Plausible Intervention Targets

The first step in our design process was to draw on a combination of psychology and adolescent social media use literatures to identity relevant psychological constructs that we could target. In what follows, we discuss the theory and prior empirical work that led us to focus on two specific factors: (1) youth self-regulation skills and (2) in-the-moment self-monitoring and goal setting.

3.1.1 Targeting Self-Regulation Skills. As discussed in Section 2.1, prior work has shown that many teens routinely experience a lack of personal agency in their social media use [20, 60, 81]. Teens who experience low levels of social media agency are more likely to report lowered subjective well-being [60]. This finding is consistent with self-determination theory (SDT), which underscores the relationship between personal agency (also referred to as autonomy) and well-being [21, 70]. Therefore, our overarching goal in this work is to support adolescents' sense of agency in their social media experiences.

A sub-theory of SDT is organismic integration theory (OIT), which presents a continuum of motivation and regulated behavior [14]. According to OIT, the more self-regulated a behavior is, the more self-determined it is felt to be. OIT highlights the central role that self-regulation plays in promoting a sense of autonomy and agency in one's actions. Other theories similarly portray self-regulation as the basis for purposeful action and a key ingredient in personal well-being [4, 7, 51]. Recognizing this connection between self-regulation and well-being, other HCI researchers encourage the use of design mechanisms that target self-regulation skills when designing interventions to support digital well-being ([45]).

As discussed in Section 2.1, self-regulation is a particularly suitable focus for our target population of adolescents in light of the developmental challenges that adolescents face in self-regulating their behavior [51]. Prior research implicates self-regulation challenges in adolescents' problematic social media use (PSMU) [3, 5, 22, 37], in addition to more benign social media experiences that may not rise to the level of PSMU but are nevertheless experienced by adolescents as unsatisfying [60, 81].

Therefore, our first aim was to envision designs that would target the self-regulation skills that adolescents are most likely to disregard when faced with an immediately gratifying social media stimulus.

3.1.2 In-the-Moment Self-Monitoring and Goal-Setting Intervention. In describing his social cognitive theory of self-regulation, Bandura [4] observed that self-monitoring close in time to the target behavior is key to developing a person's self-regulation skills. This perspective is consistent with digital self-control interventions inspired by dual-process theories of cognition [47, 84]. Dual-process theories distinguish between two types of cognitive processing: Type 1 processes are intuitive, unconscious, and automatic, whereas Type 2 processes are reflective, conscious, and controlled [25]. Technology interventions that seek to change people's behavior target Type 2 processes at the moment when Type 1 processes are typically dominant. In the case of smartphone overuse, for instance, just-in-time interventions [53] might present notifications, reminders, or prompts as a person is using their phone in an effort to stimulate Type 2 processes of cognition and prevent phone overuse (see [47, 84]).

Based on this theoretical work, our second aim was to target Type 2 processes of cognition by envisioning approaches to prompt adolescents' self-monitoring and goal setting prior to opening a social media app on their phone. Our goal is to help adolescents reframe the ensuing social media experience so that it feels intentional and purposeful rather than reflexive and devoid of meaning.

3.2 Design of the Preliminary App Prototype

Drawing on the theories outlined in Section 3.1, we identified a set of interaction design choices for the design of the preliminary Locus app prototype. First, we designed Locus as a *wrapper application* that adolescents use to open the social media apps on their phone. This design choice allows us to shape users' entry experiences into their social media apps.

Second, we generated an initial set of *app-entry prompts* that appear the first time a teen opens a social media app on any given day. Before being taken to their desired app, adolescents are prompted to identify and reflect on their goals and intentions for their social media session. In this way, the app-entry prompts encourage adolescents to engage Type 2 processes of cognition at the moment when their Type 1 processes are typically strongest. Beyond Type 2 engagement, the prompts are aimed at helping adolescents to frame the ensuing social media experience in personally meaningful terms.

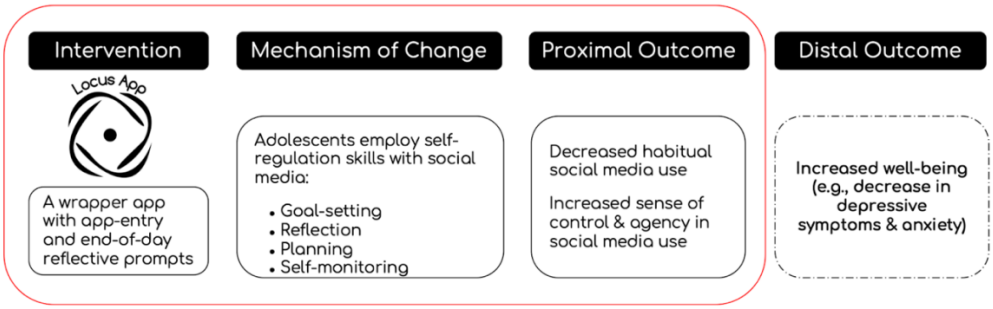


Figure 1: Theory of change model, outlining intervention, mechanism of change, proximal outcome, and distal outcome. The distal outcome is not a focus of the current investigation, which focuses on the intervention design and its acceptability and potential effectiveness among adolescents.

Third, we identified a set of *general end-of-day prompts* that encourage users to reflect on and self-monitor their social media use during the course of the day, as well as identify strategies and set goals for the following day. These prompts are not designed specifically to override Type 1 processes in-the-moment, but rather to help adolescents direct their Type 2 processes toward their social media use by engaging and exercising key self-regulation skills.

For both types of prompt (app-entry and general end-of-day), users can respond either through text entry or voice recording. They can also choose to dismiss the prompt altogether and proceed directly to their intended app. Drawing on prior research [46, 84], we made these design choices in an effort to strike a balance between being overly restrictive (a danger in requiring a response) and not engaging enough (a danger in prompts that do not invite a user response). Even if an adolescent chooses not to record a response, they will still see the prompt and must actively dismiss it in order to enter the desired social media app. Thus, adolescents are still receiving the intervention even if they do not record a response.

Figure 1 presents the theory of change associated with our intervention. We hypothesize that the interaction techniques described in this section (*intervention*) will support adolescents' social media-related self-regulation behaviors (*mechanism of change*), leading to a decrease in habitual use and—over time—an associated increased sense of control and agency in their social media use (*proximal outcome*). Because of the links between meaningful technology use and well-being [17, 42, 60], we would expect to see eventual increases in adolescents' personal well-being, such as lowered depressive symptoms and anxiety levels, if used consistently over a longer period (*distal outcome*).

3.3 Co-Design with Teens to Update and Finalize the Prototype

We conducted virtual co-design sessions with nine teens in summer 2021 following the initial design and development of the prototype app. The goals for these sessions were to: (1) gain initial insight into how Locus might be taken up in the context of teens' daily social media use; (2) solicit feedback on the app's core features; and (3) identify and fix bugs in the app to improve its usability.

Following approval from our university's institutional review board, we recruited teen participants through word of mouth, social media announcements, and our university's research-participant pool. We divided the nine participants we recruited into two groups. The first group included four teens aged 14-16 years (M_{age} = 15.0 years; 2 male, 1 female, 1 non-binary; 2 Asian, 2 white). The second group included five teens aged 15-18 years (M_{age} = 16.2 years; 3 male, 2 female; 3 Asian, 2 white).

Both groups of participants took part in three co-design sessions each lasting 2.5 hours. These sessions were structured similarly following best practices for co-designing with teens [6, 31, 65, 85]. All sessions took place via Zoom and used Miro boards to engage participants in different forms of hands-on ideation. The sessions were led by three PhD student researchers with extensive experience in co-design and working with teens in a participatory manner.

We used teen participants' feedback to update the app's prompts before deployment. In particular, we attempted to use a more casual, colloquial tone that would be appealing to teens, and we varied the phrasing of the prompts (e.g., by using a mix of questions and statements).

Teens provided inconsistent feedback on the timing of the prompts. Some teens wanted greater frequency, whereas others wanted less. Some teens appreciated receiving the questions before entering a social media app, whereas others were frustrated by having to delay their entry. We decided on one app-entry prompt per app per day and one general end-of-day reflection prompt, with the goal of encouraging teens to engage in consistent reflection while not slowing them down so much that they become annoyed and stop using Locus altogether.

During user testing, we learned that most teen participants entered social media platforms directly from the notifications that appeared on their phone's home screen. As a result, these teens rarely entered their social media apps through Locus and therefore never encountered the app-entry prompts. We addressed this issue by programming Locus to control all social media notifications so that clicking on a notification would take the user to Locus first (see Figure 2A) instead of directly to the desired social media app.

Additional bugs that we identified and fixed included incorrectly-timed notifications and recurring notifications that would not go away, a misaligned text box and questions, and lags in loading the prompts and social media apps.

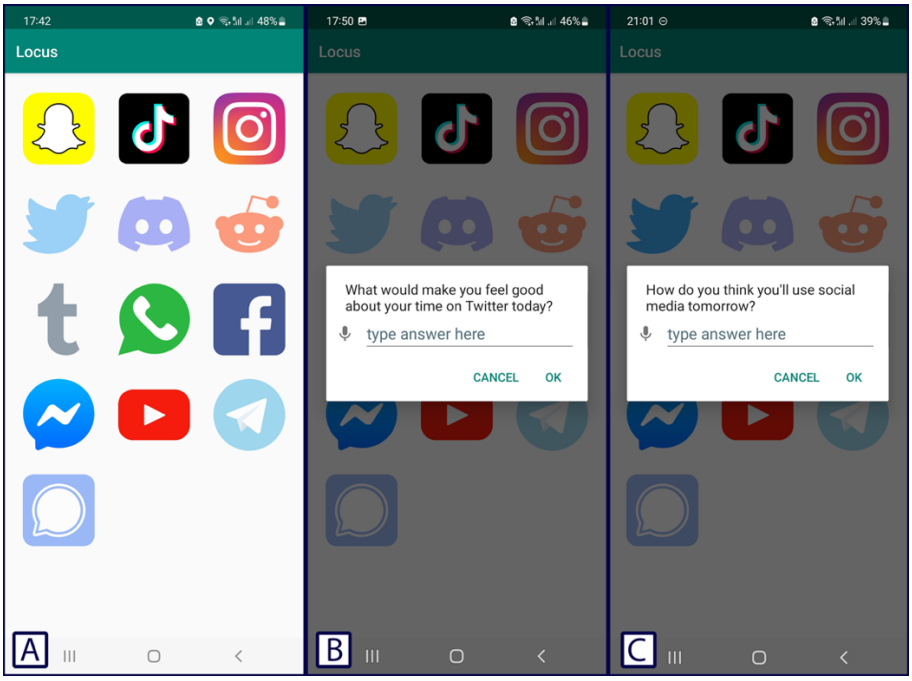


Figure 2: Screenshots of the Locus app. A: The application landing page where users access their social media apps; B: An app-entry prompt that appears after clicking on a social media app; C: A general end-of-day prompt received every day at 9pm.

3.4 Final Locus Application Taken to Field Deployment

Locus is a mobile application available on the Google Play Store for Android devices with an Android version above 7.0 (Nougat). It is designed as a wrapper application that allows users to open social media apps directly through the Locus app. After opening Locus on their phone, users see all of their social media apps displayed together (see Figure 2A). When they select an app such as TikTok or Snapchat, they are shown a text-based reflective prompt before being taken to the desired app (see Figure 2B). Entering a response is optional and can be done via text or speech recording. The app-entry prompts are programmed to appear upon the first daily use of a social media app. Users see one prompt per day per app. Locus also sends a general prompt in the form of a notification once per day at 9pm (see Figure 2C).

The backend of Locus consists of a Python server that provides the list of prompts and stores users' responses to prompts. This means that the prompts can be changed at any time without having to update Locus on users' phones. In addition, the server authenticates registered users with an email and password combination and hosts a website with instructions on how to install and use Locus.

4 OPEN TRIAL FIELD DEPLOYMENT

We conducted an open trial field deployment (N=54) to examine the acceptability of Locus and how the app was taken up and experienced by adolescents in their everyday lives. We sought to answer the following research questions:

RQ1: How do teens interact with the Locus app features over the course of the two-week deployment period?

RQ2: To what extent does Locus engage teens in goal setting, self-monitoring, and other self-regulation behaviors related to their social media use?

RQ3: Does using the Locus app help teens experience greater self-control and agency in their social media use?

4.1 Participants

Following approval from our university's institutional review board, we recruited English-speaking participants from the United States through a paid Instagram ad that included a link to an initial screener survey. To be included in the study, prospective participants had to live in the Unites States, own and use an Android phone, be between the ages of 14 and 18, have a reliable internet connection, and express an interest in changing their approach to social media.

Sixty-six participants met the inclusion criteria and successfully installed the Locus app on their phone. Of those participants, 54 completed all study procedures, and 26 participated in a post-study interview (see Section 4.2). Table 1 includes a summary of participant demographics.

4.2 Procedure

Participants who passed the screener were taken directly to a pre-deployment survey (via Qualtrics) that concluded with a link to the tutorial for installing and using Locus. At the end of two weeks using Locus, we asked participants to complete a post-deployment survey (via Qualtrics) and invited them to participate in a follow-up

Table 1: Participant demographics

Participant Demographic	Percentage (N = 54)
Age (years)	15yo: 20.37% (11), 16yo: 37.03% (20), 17yo: 41.74% (22), 18yo: 1.85% (1)
Subjective Social Status [a]	Family: M = 6.68 (SD = 1.91), Self: M = 7.25 (SD = 1.69)
Race/Ethnicity [b]	Black: 53.7% (29), White: 35.1% (19), Asian: 14.8% (8), Hispanic or Latine: 3.7% (2), Other or mixed ethnicity: 1.85% (1)
Gender [b]	Female: 24.5% (13), Male: 75.9% (41)

[a] MacArthur Scale of Subjective Social Status—a two-item instrument that measures how a young person perceives their family's (Family) and their own (Self) social standing. For each item, respondents provide a rating from 1 (very low) to 10 (very high) [27]

[b] Participants were able to select multiple options

interview via Zoom. Participants were given a $25 gift card upon completing the study.

The pre-deployment survey asked questions about participants' social media use, including which apps they used most frequently, how they felt about their use of these apps, and how they felt about their social media use more generally. In addition, we included three validated scales measuring self-control and agency in social media use:

- **Self-control of social media use.** Following Lyngs and Lukoff [45], we adapted a digital self-control version of the Brief Self-Control Scale (BSCS) [76] to measure self-control ability in the context of adolescents' social media use (e.g., *"In the past week, I was good at resisting temptation on social media"*). Participants used a 5-point Likert scale to indicate how much each of the 13 statements reflected their experience using social media in the past week (1=not at all; 5=very much). The internal consistency of the scale was satisfactory (Cronbach's alpha =0.77).
- **Absentminded social media use.** We adapted the 10-item Smartphone Use Questionnaire-Absentminded (SUQ-A), which measures perceptions of absentminded smartphone use [50], to measure participants' levels of absentminded social media use. We chose absentmindedness because of its opposition to intentionality, hypothesizing that the Locus app would decrease participants' absentminded social media use as they became more intentional in their approach to social media. Participants used a 7-point Likert scale to indicate how frequently they had the experiences identified in each statement (e.g., *"How often do you check social media out of habit?"*) on a typical day (1=never; 7=always). The internal consistency of the scale was satisfactory (Cronbach's alpha =0.89).
- **Autonomy in social media use.** We adapted the 4-item autonomy sub-scale of the Technology Effects on Need Satisfaction in Life (TENS-Life) [58]. Drawing on self-determination theory (SDT) [21], this scale measures the extent to which people believe a technology has had an impact on their psychological need for autonomy (a construct that is closely aligned and often interchangeable with agency). Peters et al. [58] instruct that the TENS-Life can be adapted for any technology or platform. Our social media adaptation asked participants to think about the social media platforms they use most frequently and rate how they have affected them (1=not at all; 5=very much). Each item of the sub-scale is reverse-scored (e.g., *"I feel pressured to use social media more often than I'd like."*). The internal consistency of the scale was satisfactory (Cronbach's alpha =0.75).

The post-deployment survey mirrored the questions from the pre-deployment survey, with the addition of targeted questions about participants' experiences using the Locus app, how consistently they used it, and any bugs they encountered. To assess the app's self-perceived impact on social media use, participants were asked how strongly they agreed or disagreed with the following statements: *"The Locus app helped me to think about my goals for using social media"* and *"The Locus app made me feel more in control of my social media use."* Response choices were on a Likert scale from 1 (strongly disagree) to 5 (strongly agree).

At the end of the two weeks, 26 participants took part in a follow-up interview. Interview questions included both user experience questions and questions about the Locus app's self-perceived impact on participants' approach to social media. The interviews gave us a qualitative understanding of how teens appropriated and experienced Locus and implications for further redesign.

To analyze the interviews, we used a reflexive thematic analysis (TA) approach that emphasizes researcher reflexivity and interpretive depth (in contrast to post-positivist approaches that rely on inter-coder agreement) [11–13]. Drawing on theory and prior research (see Section 2), and guided by our three research questions, the first author read through the interview transcripts and interviewer notes multiple times, highlighting and grouping ideas that addressed each research question. For every highlighted idea and across each interview transcript, the researcher recorded instances of both confirming and disconfirming evidence. The first and third authors used this analysis to frame a series of interpretive conversations and construct themes.

5 RESULTS

The results of the deployment study are organized around our three research questions, which address teens' interactions with the Locus app's features (**RQ1**); engagement in self-regulation behaviors (e.g., goal setting, reflection, self-monitoring) (**RQ2**); and perceived changes in teens' experiences of self-control and agency in their social media use (**RQ3**). To protect participants' privacy, we use numerical participant identifiers in place of actual names.

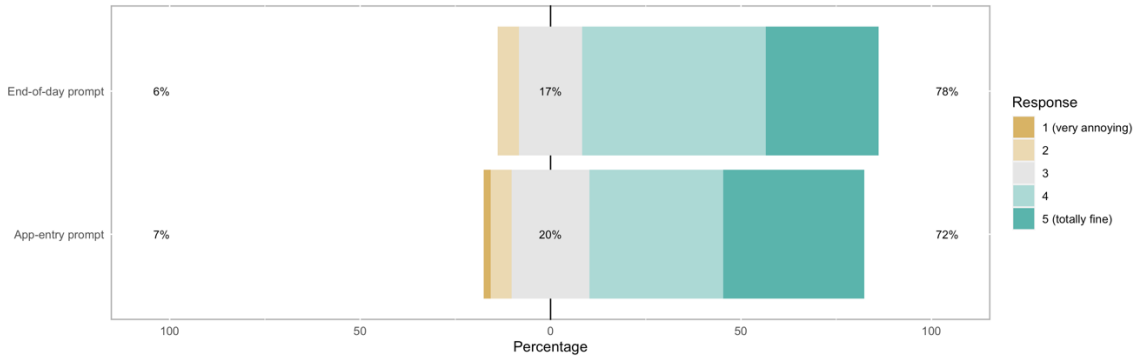


Figure 3: Distribution of responses to the post-deployment survey question asking participants to rate how annoying they found it to receive the end-of-day and app-entry prompts (*N*=54).

5.1 RQ1 – Participants' Interactions with the Locus App's Features (Acceptability)

Our first research question focused on the acceptability of the Locus app to adolescent users. In designing Locus as a wrapper application, we were initially concerned that teens would find it cumbersome or annoying to hide the social media icons on their phones and enter their desired apps instead through Locus. Although some participants said they had to remind themselves initially to go through Locus, participants said they adapted quickly and found it straightforward to access their social media apps in this way. 80% of participants reported that they opened their social media apps through the Locus app most of the time or all of the time during the course of the two-week study. Of the 20% who used Locus less regularly, they usually said they either forgot or found it easier to access their social media apps directly instead of being slowed down by an intermediary app. P51 (male, age 16) explained: *"Sometimes if I really needed to get to an app, I just kind of went to that directly, because it was just easier to do that in the moment."*

We asked participants to reflect on and rate how they experienced receiving each type of prompt. Figure 3 shows the distribution of their responses on a 5-point Likert scale ranging from *very annoying* (1) to *totally fine* (5). Very few participants rated either prompt as *very annoying*, with the app-entry prompts receiving a slightly higher proportion of such ratings than the general end-of-day prompts. Reflecting this general lack of user frustration, P16 (male, age 17) said of the app-entry prompts: *"Having that little bit of extra friction was not. . .it wasn't as annoying as I thought it was gonna be. And actually, it was more helpful too, in that it made me reflect."*

The frequency of the prompts and the ability to dismiss them without recording a response both appeared to play a role in minimizing user frustration, which is critical to the acceptability of the intervention. Participants tended to agree that the number of prompts they received each day did not feel onerous. Some participants cautioned that increasing the number of prompts (both app-entry and end-of-day) may lead to people abandoning the app out of frustration. Others felt that increasing the frequency slightly would encourage people to reflect more on their social media use. P61 (male, age 16) addressed this tension: *"I feel like if you asked a question every single time [you opened a social media app], it'd be annoying, but also, it would make you think about [your goals for social media] every single time."*

Several participants appreciated that responding to the prompts was optional. P35 (female, age 17) observed: *"The app is fairly non-invasive. . .it doesn't block you every time you get a notification. I think that would have been too invasive for me. . .but it was enough so that you're still aware of things."* However, some participants still felt compelled to record a response even though it was optional. P61(male, age 16) explained:

> *"Sometimes it was slightly annoying to use because I just needed to go, get into the app. . .I was like, 'I don't really wanna answer the question right now.' But I was kinda forced to either do it now or it was like, not answer at all."*

This participant started to experience the prompts as a *"barrier to entry."* He suggested that he would have liked the option to postpone answering a prompt until a time when he was in the right frame of mind rather than have the prompt disappear altogether. Several other participants similarly reflected that they would like the ability to customize the app, such as by controlling the timing of the prompts, deciding when to answer them, selecting the prompts they liked best, adding their own prompts, and being shown their responses over time.

5.2 RQ2 – Engagement in Self-Regulation Behaviors Related to Social Media Use

To determine the extent to which participants perceived the intervention as having impacted their self-regulation behaviors, we analyzed their responses to interview questions about whether and how Locus changed their approach (both beliefs and behaviors) to social media. We also analyzed responses to a Likert-scale question included on the post-survey that asked participants to rate how much they agreed or disagreed that Locus helped them to think about their goals for using social media.

81% of participants agreed or strongly agreed that Locus helped them to think about their goals for using social media (see Figure 4). P16 (male, age 17) observed: *"[Locus] helps me to. . .think about what I'm going to do on Instagram or whatever social media I'm using and*

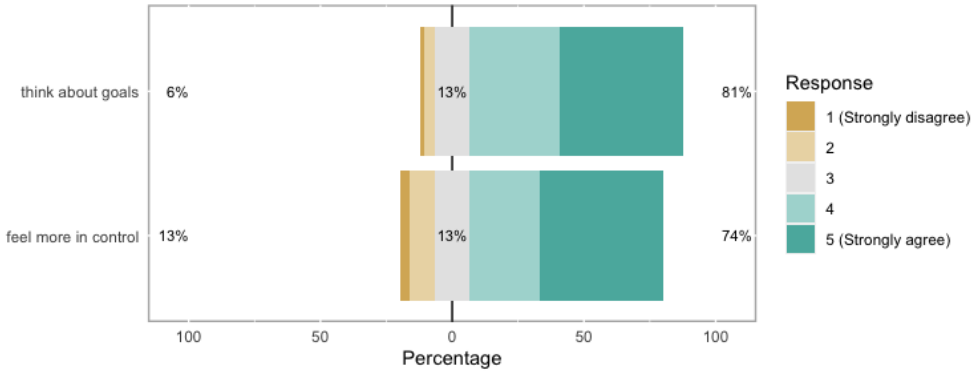


Figure 4: Distribution of participants' responses when asked if Locus helped them think about their goals and feel more in control.

kind of have that goal in mind." Some participants, like P51 (male, age 16), connected goal setting with clarifying the value they were getting from their social media use:

> *"I thought of why am I going on this social media app and what's the benefit for me? Most of the time it was not great, so I just kind of abandoned [social media] and get [sic] to do more important things."*

In the follow-up interviews, participants' observations extended beyond goal-setting to include other self-regulation behaviors such as self-monitoring and reflection. Both types of prompts—app-entry and end-of-day—contributed to these self-regulation behaviors. The end-of-day prompts helped participants to reflect on the day's social media use generally and then use those reflections to set goals for the following day: *"I got time to reflect and think about what I did on social media and know if I achieved the right goals or I was just wasting time on social media and watching videos and having my laughs"* (P58, female, age 17). The app-entry prompts were more targeted than the end-of-day prompts and helped participants to focus on the immediate social media session before them. P16 (male, age 17) observed: *"I felt like having a pause before I actually went in [to a social media app], definitely made me reflect a little bit before I went on to do what I was doing."*

Participants' baseline stance toward social media and motivations for enrolling in the study seemed to affect the value they received from the Locus prompts. P28 (male, age 18) came to the study motivated to change his social media practices:

> *"When we began this whole study, my goal was to use social media less than the way I used it, because I thought I was so addicted to social media. . .and checking every single notification that pops up on my phone. So Locus has a way of making me think of why I want to access the app and how long I want to stay on the app."*

In contrast, P06 (male, age 17) did not see the point in identifying goals for his social media use: *"It's a social media app. It's not meant for anything else besides time passing and finding ways to connect to other people. So I can't really think that question [about goal setting] really helps too much."*

5.3 RQ3 – Perceived Impact on Teens' Sense of Control and Agency in their Social Media Experiences

To examine evidence of the intervention's potential effectiveness, we conducted paired-sample t-tests to compare levels of our three outcome measures before and after the two-week intervention. Shapiro tests indicated that none of the measures violated the assumption of normality. Figure 5 shows the pre- and post-deployment distributions, means, and confidence intervals for the three measures: perceived self-control (BSCS), absentmindedness (SUQ-A), and autonomy (TENS-Life) in relation to social media use.

For all three measures, there was a statistically significant difference between condition means. On average, participants reported increased self-control ($t(52)=5.79$, $p< 0.001$), decreased absentmindedness ($t(52)=2.74$, $p<0.01$), and increased autonomy ($t(52)=5.12$, $p<0.001$) in relation to their social media use from pre- to post-deployment. Table 2 summarizes these results, including effect sizes.

In addition, 74% of participants agreed or strongly agreed with the statement *"The Locus app helped me feel more in control of my social media use"* (Figure 4). Taken together, these results provide preliminary evidence that the Locus intervention had a positive impact, at least in the short-term, on participants' perceived self-control and agency in their social media use. In the remainder of this section, we dig deeper into these aggregate trends by drawing on participants' individual interview responses.

For some participants, their perceived increased sense of control came from being able to reduce the amount of time they were spending on social media: *"I learned basically self-control. I learned that, okay, you would have to minimize time you spend on social media and focus more on work and other things"* (P58, female, age 17).

For other participants, their perceived increased sense of control had less to do with the *amount* of time they used social media and more with the *quality* of their interactions. A common theme in participants' descriptions of their social media use before the study was their tendency to use social media "mindlessly" and out of habit. For some, Locus helped them to disrupt their habits, at

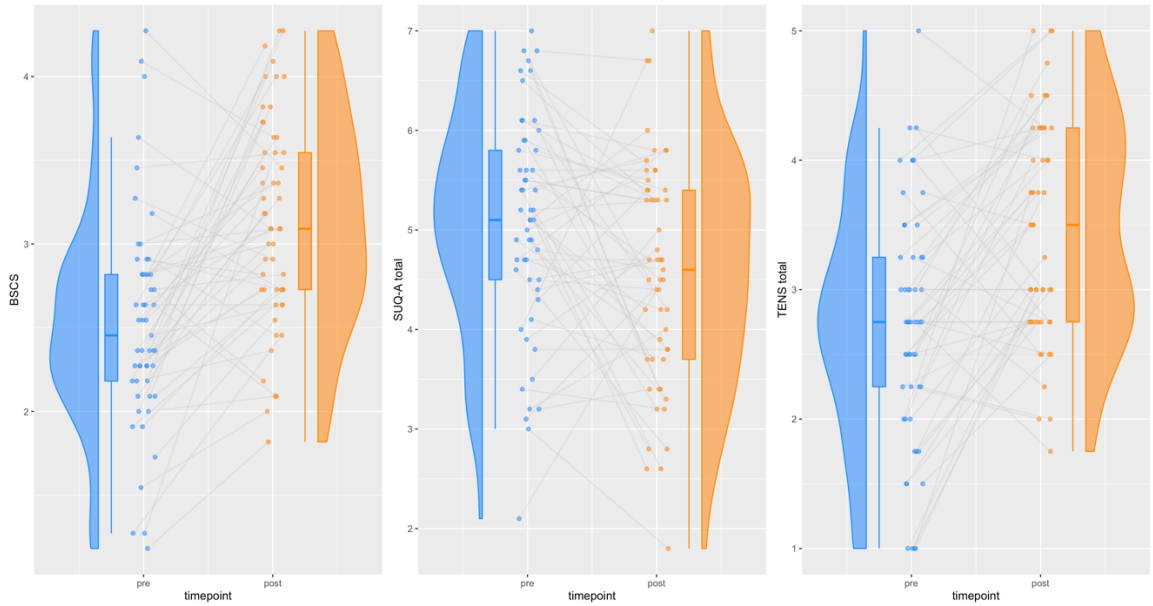


Figure 5: Raincloud plots showing distributions of the raw data, means, and confidence intervals for the three pre- and post-deployment measures used in the field deployment (*N*=53). Left to right: BSCS, SUQ-A, TENS-Life.

Table 2: Differences between pre- and post-deployment on self-control, absentmindedness, and autonomy in SM use (*N*=53)*

	Pre-Deployment		Post-Deployment				
	M	***SD***	***M***	***SD***	***t(52)***	***p***	**Cohen's d**
Perceived self-control (BSCS)	2.53	0.64	3.13	0.62	5.79	<0.001	- 0.79
Absentmindedness (SUQ-A)	5.05	1.11	4.56	1.14	2.74	<0.01	0.38
Autonomy (TENS-Life)	2.72	0.92	3.45	0.85	5.12	<0.001	- 0.70

*1 participant had incomplete data on these measures and was excluded from the analysis.

least during the two weeks they used the app. P06 (male, age 17) explained: *"During the night at like 2AM or something, where you'd be mindlessly scrolling through Instagram Reels or TikToks. . .it's nice to have some sort of thing that's holding you back from wasting your time."*

This effect did not apply to all participants, however. P61 (male, age 16) explained that, over the course of the two weeks, responding to the Locus prompts itself became as mindless and habitual as his social media use: *"After a little while, I just kinda, like, auto-pilot did the questions."* Unsurprisingly, this participant did not feel that Locus helped him to feel more in control of his social media use. Although he said he engaged in more reflection than previously, his relationship to social media overall remained unchanged after the intervention.

Several participants tied their perceived increased sense of control with a greater intentionality in their social media interactions. Locus helped P16 (male, age 17) to clarify his intentions for opening social media:

> *"It was really nice to think before I opened a social media app, just because normally it's kind of like an impulse thing like, 'Oh, I'm bored, I'm gonna go on Instagram or whatever.' But now I have to be. . .like actually think about what I'm going to do, why I'm on here."*

This participant went on to observe that he was now less likely to engage in social comparison when scrolling through sites like Instagram, which had previously been a source of stress for him. By the end of the study, he was feeling more satisfied with his social media use now that he was using it in a more intentional way.

6 DISCUSSION

6.1 Extending Digital Well-Being Designs to an Adolescent Population

Adolescents are an under-researched group in HCI interventions that target digital well-being even though adolescents struggle to control, find meaning, and experience agency in their technology use [60, 68, 81]. Our work provides preliminary evidence that

interaction designs that prompt teens to engage in goal setting, reflection, planning, and self-monitoring are an acceptable form of in-the-moment intervention and can potentially help teens—at least in the short-term—to use social media in a more intentional way. Results from our open trial field deployment (N=54) showed that, on average, teen participants reported perceived increased self-control and autonomy and decreased absentmindedness in their social media use over the two week-period of using Locus. The results provide a basis for future work investigating adolescents' long-term use of and response to digital interventions, such as Locus, that provide adolescents with in-the-moment support for self-regulating their social media use.

Our exploratory field deployment provided evidence supporting the initial steps in our theory of change (Figure 1). Interview participants discussed how Locus (*intervention*) prompted them to engage in self-regulation behaviors related to their social media use (*mechanism of change*). Over the course of the study, participants observed themselves engaging in less habitual social media use (characterized by "mindless scrolling"), approaching social media with greater intention, and experiencing more meaning in their interactions (*proximal outcome*).

We would not expect—and therefore did not measure—any long-term changes in well-being (*distal outcome*) over the course of just two weeks. Future work is needed to examine the long-term appropriation of Locus, including changes to how youth engage with the app over time, and whether it succeeds in influencing their well-being. Such an investigation is outside the scope of the current work, which focused on the initial design of the Locus intervention and an exploratory investigation into its acceptability and potential effectiveness with a group of adolescents.

The current study connects to and extends prior HCI research on digital well-being (e.g., [29, 44, 48, 68, 77, 84, 86]). We applied insights generated from existing work to a population that is distinctly positioned to benefit from design interventions targeting digital well-being. Adolescence represents a sensitive period for the development of self-regulation skills [51]; it is also a time of intensive social media use, which many youth struggle to manage in an intentional way [20, 81]. In light of these challenges, we see an opportunity for HCI research to identify interaction designs that help build the self-regulation skills and shape the social media habits that adolescents will carry with them into adulthood. The current study contributes insight into promising interaction designs that support adolescents' intentional social media use, as well as the contexts and individual characteristics that shape how they are taken up by specific adolescents.

The results of our deployment study suggest that the set of interaction designs employed by Locus (see Section 3.4 and Figure 2) are acceptable to adolescents and hold promise for supporting intentional social media use in an adolescent population. Overall, adolescent participants adopted and used the Locus app with little trouble or annoyance. There was a general consensus among interview participants that the frequency of the prompts (one app-entry prompt per app per day, one general, end-of-day prompt) was not too great to be annoying. In light of prior work showing high rates of user frustration and abandonment of self-control tools [43, 46], striking a balance between user engagement and frustration is critical. The inclusion of a response entry option (as opposed to requiring users to record a response to every prompt) succeeded at engaging participants in the target self-regulation skills without producing too much frustration due to the fact that it was optional. Overall, it appeared that Locus came close to a "Goldilocks" level of self-control support for most participants [46].

6.2 Accounting for Individual Variation when Designing for Adolescent Digital Well-Being

We gained useful insight from the few participants who said they found little value in using Locus. For instance, P61 (male, age 16) observed that, rather than disrupting his social media habits, answering the app-entry prompts themselves became as mindless and habitual as his social media interactions. Although a minority view, this perspective points to the need for future work to explore which adolescents specifically are most likely to benefit from design interventions, such as Locus, that target self-regulation skills in the context of social media use.

For example, prior research suggests that problematic social media use may be higher among adolescent girls than boys [37]. Future work could explore the moderating role of demographic differences such as gender, age, socioeconomic status, and race, as well as the role of individual characteristics such as executive function, personality type, levels of existing dissatisfaction with social media, and motivation to change one's social media patterns.

Even among the majority of participants who perceived benefits from using Locus, several of them identified changes they felt would improve the user experience and the intervention's effectiveness. Moreover, there was considerable variation in their feedback. Some participants wanted more frequent prompts, others less. Some wanted it to be mandatory to respond to the prompts, others said responding already felt mandatory. Some participants wanted to be able to customize the app, for instance, by selecting their favorite prompts and deciding for themselves when to answer them. Several participants offered ideas for new features, such as a visualization tool that shows their responses to prompts over time.

Returning to the concept of a "Goldilocks" level of support [46], these findings indicate considerable individual variation across adolescents in the "just right" level of support that is most likely to be effective in scaffolding their self-regulation behaviors. It is also worth exploring designs that are tailored to the environments of specific social media platforms (e.g., [86]) and the distinct ways that adolescents engage with them. Whereas social comparison may be a relevant target of intervention for platforms like Instagram, the lure of auto-play may be a more relevant focus for sites like YouTube and TikTok (see [20]). Therefore, we recommend further design investigations before testing the long-term impacts of Locus and other interventions targeting adolescents' intentional social media use.

7 CONCLUSION

Digital well-being is a rapidly growing area of research in HCI. There has been relatively little work to date that addresses adolescents as a specific user group, despite the fact that many adolescents struggle to manage and find meaning in their digital media experiences. In light of their developmental challenges with self-regulation, adolescents stand to benefit from interventions that are

designed for and with them from the start. Drawing on psychology theory and co-design practices, we developed Locus, a mobile application that shapes users' entry experiences into the social media platforms on their phones and encourages end-of-day reflection about their interactions. Results from an open trial field deployment (N = 54; M_{age}=16.2 years) showed that Locus was an acceptable intervention for this group of adolescents and showed evidence of potential effectiveness. Participants reported engaging in more frequent self-regulation behaviors in relation to their social media use and perceived increases in their self-control and autonomy and decreases in their absentminded social media use. The results from the current study point to opportunities for future research, such as customization features that allow adolescents to tailor the intervention for their individual needs.

8 SELECTION AND PARTICIPATION OF CHILDREN

All procedures involving minor participants received approval from our university's institutional review board (IRB). We recruited participants through word of mouth, social media announcements and advertisements, and our university's research-participant pool. All procedures were explained to potential participants prior to their participation, including the voluntary nature of participation, the ability to end participation at any point during the study, and the procedures in place to assure the confidentiality of their participation. We shared this information with participants electronically, and they indicated their assent by clicking on a button indicating they understood the nature of the study and agreed to participate in it. For those teens who participated in co-design sessions and follow-up interviews in the deployment study, we shared study information both electronically and verbally. The university IRB designated the study minimal risk and waived the requirement for parental consent.

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